SHRAYAS RAGHUNATHAN

Software Engineer | Java | Spring Boot | Distributed Systems | Architecture | AI/ML

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PROFESSIONAL SUMMARY

Software Engineer with **4+ years** of experience at Citi building, maintaining and modernizing **Java/Spring Boot** microservices for consumer and commercial Retail Cards acquisition platforms. Experienced in **distributed systems, API design**, concurrency, resiliency, performance engineering, production incident resolution, and **cloud migration architecture**. Strong research background in **Reinforcement Learning**, Deep Learning, NLP, and embedded systems, including **peer-reviewed research and patented engineering work**.

TECHNICAL SKILLS

Languages & Backend: Java, Python, SQL, JavaScript, Spring Boot, REST APIs

AI / ML: Reinforcement Learning, Deep Learning, NLP, Generative AI, MLOps, PyTorch

Architecture & Design: Microservices, API Design, HLD, LLD, Design Patterns, SOLID, Distributed Systems

Concurrency & Resilience: Multithreading, Concurrent Processing, Service Discovery, Eureka, Caching, Circuit Breaker, TimeLimiter

Cloud & DevOps: AWS, OpenShift, Jenkins, Bitbucket, APIM, CI/CD

Engineering Practices: Agile/Scrum, Jira, SDLC, Defect Management, Code Quality, SonarQube

Data & Performance: MongoDB, GemFire, JMeter, Load/Stress Testing, Profiling, Splunk, Production Debugging

Security & Frontend: Authentication, RBAC, TLS, AES-256, Angular, CSS, Figma, Adobe Analytics

PROFESSIONAL DEVELOPMENT

System Design: System Design Masterclass — FAANG 2026 — **Udemy**
Low-Level Design: Design Patterns & SOLID Principles — **Udemy**
Deep Learning: Introduction to Deep Learning with PyTorch — **Udacity**
Reinforcement Learning: Prediction & Control with Function Approximation; Fundamentals of Reinforcement Learning — **Coursera**

LEADERSHIP & ACHIEVEMENTS

- Recognized by the **Head of USPB Digital** with a Certificate of Appreciation for **outstanding performance**, June 2026.
- **Highest team performance rating** for **3 consecutive years**, with multiple Best Sprint Performer awards.
- **Honorable** Mention — Citi **AI-OPS** Hackathon 2023, developing a Splunk-based solution to identify CRS submission failures and **automate retry**-data population.
- **Winner** — Indo Data Week: Climate Data Hackathon 2022, developing an ML/RL-based crop recommendation framework using multivariate climate and meteorological data.(Medium Article)
- **Chairman** / **Treasurer** — MIT Robotics Association: led DEXBOT'22, a National-Level Technical Symposium, securing **₹50K in corporate sponsorships** and establishing a cross-disciplinary mentor team for **Robotics and AI research**.

PROFESSIONAL EXPERIENCE

Citi — Software Engineer *Jul 2022 – Present | Chennai, India*

2026 — Consumer Credit Servicing & Platform Modernization

- Expanded Consumer Credit Servicing capabilities by delivering **paperless enrollment for Best Buy**, dynamic **customer alert enrollment in pre-registration flow**, and registration-flow UI screens for card-management journeys.
- Resolved a **production defect** in paperless enrollment where enabling email unintentionally **de-enrolled SMS** regardless of prior customer preference; corrected the **preference-update logic** to preserve existing channel selections.
- Designed the target **architecture** to sunset Cheetah-based email triggering and onboard **Salesforce Marketing Cloud,** introducing **Kafka**-driven eventing alongside FusionDB; implemented an **Async MDC Executor** for tiny-URL generation, FusionDB payload construction, and **Kafka** publishing, while retaining FusionDB as the source of truth during migration. Focused on maintaining **consistency** across DBs.

2025 — Architecture, Distributed Systems & Optimization

- Architected and executed a major Commercial Acquisition design migration across **APIM**, Platform Security Gateway and 4 **microservices**, replacing **manual** routing with **Netflix Eureka service discovery**; reduced API **latency** by **~1 second** and established a **plug-and-play** architecture enabling subsequent AWS migration.
- Eliminated redundant **service-to-service rule-configuration** calls via a **Singleton**-based configuration cache with **deep-copy protection**, reducing initial page-load **latency** by **30%**, improving **API responses** by up to **42%**, and eliminating **15–20 MB memory overhead** per microservice.
- Architected and developed a **Release Planner** application with 15+ APIs, optimizing retrieval of **25K+ entries using parallel fetching**, **caching**, debouncing and pagination, reducing **response time** from **~2 minutes to <20 ms**.
- Designed a **unified responsive UI** for Macy's and Bloomingdale's consumer credit applications, replacing separate desktop/mobile implementations with a single **template** with **CSS specificity override** and reducing future development effort by **~50% across 10+ pages**
- **Supported OpenShift-to-AWS migration** by resolving cloud-transition issues and improving production observability through **structured Kibana logging**; served as **one of two engineers authorized for production deployments**, owning controlled releases and production fixes.

2024 — Platform Modernization, Resiliency & Performance

- DB Migration for 4 Commercial Acquisition **microservices** from **MongoDB** to **GemFire** for low-**latency** data access, restructuring persistence behind a decoupled database interface while maintaining **DC Affinity** and data consistency.
- **Centralized** configuration for **4 microservices** by migrating **OpenShift** ConfigMaps to a Bitbucket-backed Config Server, eliminating duplicated platform **configurations** maintained at app team end.
- Led **Java 17** migration across 10+ **microservices**, resolving PowerMockito/static-mocking incompatibilities while maintaining **90%+ code coverage**; introduced **Circuit Breaker** and **TimeLimiter resiliency patterns** and deployed 2 applications to production.

- **Designed and developed secure PII/encryption services between Citi and credit-bureau systems** and contributed architecture for a **first in industry Credit Unlock capability**, using iframe-loaded bureau authentication and **KIQ validation** to temporarily unlock accounts and allow eligible customers to resume credit approval.
- Led VPE(Virtual Performance Engineering) across 6 sprints, executing **load**, **stress** and **profiling** tests to secure **performance certification**; automated encryption/decryption test flows and delivered Bureau Unlock performance validation within 48 hours using service mocks.
- Identified and eliminated **cross-microservice data-model inconsistencies** in DEV, preventing **downstream** override, OTP and KIQ failures before production deployment.
- Strengthened **OfferChoice security through response-code-driven role elevation**, replacing **hardcoded CN/tenant** authorization with **role-based access controls** and removing unused TLS certificates.

2023 — Product Engineering & CitiPay

- Designed and developed **4 Spring Boot REST APIs** powering CitiPay's Digital Line of Credit (**DLOC**) and Merchant Installment Loan (**MIL**) products, supporting ~2 **TPS** average / 3 **TPS** peak per API; executed performance, component-load, profiling and breakpoint testing for production readiness.
- Designed and delivered the backend for **6 prequalified CitiPay/MIL customer journeys within 30 days**, owning end-to-end implementation under an accelerated delivery timeline.
- Implemented **Multiple Offer** onboarding for MIL, developing a **domain** and **CRUD** API integrated with the backend rules engine to support offer selection and **dynamic Loan Note Agreement** rendering.
- Integrated ChargeAfter into CitiPay flows with request-source validation to **restrict unauthorized traffic**; delivered **SMS-based DQT**(**D**ynamic **Q**R scan / click on **T**ext) support and integrated Ensighten tracking for customer interactions for analyzing and reducing dropouts.
- Architected an **ML/DevSecOps** vulnerability-remediation pipeline, integrating **GPT vulnerability recommendations** with **Jenkins** through a reusable **pipeline plugin**; built a **Splunk**-based log-processing API for the **AI-OPS** Hackathon to identify failed CRS applications and pre-populate retry data (**Honorable Mention**).

2022 — Security Engineering & Distributed Processing

- Architected a unified vulnerability intelligence dashboard aggregating vulnerability data sources Snyk, Blackduck and Checkmarx; implemented **asynchronous** service orchestration and worker-pool processing across **3 vulnerability streams** for hundreds of paginated issue-detail calls, **reducing** data-fetch time from **~6 minutes to <750 ms**; presented the POC to senior leadership, resulting in multiple team onboarding requests.
- Implemented automated bearer-token lifecycle management with scheduled renewal at the 3-month expiry boundary; secured credentials using **AES-256** encryption and decrypted tokens only during outbound authentication.

PRODUCTION INCIDENTS & RELIABILITY

- Diagnosed **100K+** erroneous requests raising **alarm** in a single day following an **Apigee migration** by correlating Splunk **IP-level traffic** and API timelines; identified **browser-tab reloads** and malformed bot POST traffic as root causes and deployed an **overnight** fix, increasing the **2xx success rate to 99.82%.**
- Identified the root cause of an **edge-case** production validation failure affecting **iOS autofill flows**, tracing the device-specific input path and correcting the validation handling.
- Resolved a production **race condition** caused by shared **static-state mutation**, replacing unsafe shared-state access with **deep-copy isolation** to prevent **concurrent request contamination** and eliminate restart-based recovery.
- Unblocked a critical production deployment by tracing **cross-DC traffic failures across** GRN, DMZ, APIM and Akamai; coordinated with the **Akamai team** and delivered the deployment **within 1 week vs. a proposed 1-month delay.**
- Optimized POS approval-email processing from **O(N) to O(1)** by replacing nested-JSON iteration with a **direct lookup** structure across 100+ credit-unit configurations.

RESEARCH & ENGINEERING

IIT Madras — R&D Intern *Oct 2021 – Jun 2022*

- Developed a **SciBERT-based transformer NLP** pipeline trained on **1.8M+** scientific abstracts from **Elsevier** and **Springer** for **electrocatalyst** discovery and activity prediction.
- Published the research in a **peer-reviewed journal** applying representation learning and NLP to materials science; the publication has **4 citations** to date.

Embedded Systems & Patented Engineering

- Precision Stepper-Based Powder Dispensing: sole software developer in a 3-member team; developed **embedded control software** for stepper-driven piston actuation, mapping **motor rotation to piston position** and powder volume with dispensing **precision of up to 0.001 g**.
- Vibration-Calibrated Powder Dispensing: developed a PID-based control algorithm using MPU-6050 3-axis vibration feedback to correlate **vibration characteristics with powder output** and dynamically regulate dispensing.

AI/ML PROJECTS

- **RL-Based Dynamic Voltage & Frequency Scaling (DVFS):** designed a novel Reinforcement Learning-based DVFS framework for performance-aware **power and thermal management** on Raspberry Pi 4B, extending DVFS beyond conventional scheduler-level optimization.
- **Robust Reinforcement Learning** — LunarLander: modified OpenAI Gym LunarLander with **variable landing positions** and **Gaussian** disturbances; trained and evaluated **DDPG and TD3** agents for robust continuous-control policy learning.
- Cardiac MRI Semantic Segmentation: developed a **CNN/U-Net** model for segmentation of cardiac structures using the ACDC dataset, evaluating performance through confusion-matrix-based classification and segmentation metrics, 87.55% average Dice, with 94.92% LV, 87.10% RV, and 80.63% myocardium scores.

EDUCATION

Madras Institute of Technology, Anna University — B.E., Electronics and Instrumentation Engineering, 2018 – 2022. **Graduated as Best Outgoing Student; CGPA 9.3.**

DAV School — Senior Secondary (2016): **10 CGPA**. Higher Secondary (2018): **97.25%**.